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Short Communication

Another view of the upper and intermediate explosion limits of a H_2–O_2 system

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ABSTRACT

In the present note we consider the two simultaneous mechanisms that are dominated by gas-phase reactions, namely the upper and intermediate explosion limits, and show that by employing the Le-Chatelier rule, a unified explosion limit is obtained in the form of a single analytical expression.

Explosion (self-ignition) is defined here [1] as the kinetic regime in which the chemical process is self-accelerated at a rapidly increasing rate due to the multiplication of reactive intermediates as a result of a chain avalanche and/or due to progressively accelerating self-heating. Explosion limits are the pressure–temperature boundaries for a specific fuel and oxidizer mixture ratio that separate the regions of slow and fast reaction. The existence of explosion limits in a closed vessel are generally explained by the competition between chain-carriers production and chain-carriers removal; these may occur in the gas-phase and on the solid surfaces (in particular the radical H and the less active radical HO_2), as follows.

The lower explosion limit (lower pressure branch) is determined by a balance between chain-carriers production mainly through gas-phase reactions and chain-carriers removal mainly on the surface (wall effect). As the pressure is raised, the former increases to a point at which surface destruction are no longer sufficient to prevent a branching explosion.

The intermediate explosion limit (medium pressure branch) is explained by a balance between chain-carriers production through gas-phase reactions and chain-carriers removal through three-body gas-phase reactions. As the pressure is raised, the latter become more dominant and above a specific pressure the rate of chain-carriers removal exceeds the rate of the chain-carriers production.

The upper explosion limit (upper pressure branch) is attributed to the higher gas-phase density that promotes

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reactions that enhance chain-carriers production. Above a specific pressure there is a rapid increase in the number of chain-carriers species.

In the present note we consider the two simultaneous mechanisms that are dominated by gas-phase reactions, namely the upper and intermediate explosion limits, and show that by employing the Le-Chatelier rule, a unified explosion limit is obtained in the form of a single analytical expression.

The thermal theory suggests that above a specific (ignition) temperature, owing to fast enough exothermic chemical reactions, the temperature rises, which is accelerating the reactions, thus liberating more heat. The process continues in a self-accelerating manner to culminate in an explosion. The exponential nature of the temperature rise stems from the basic chemistry postulation of Arrhenius through which the exothermic reactions progress. For that reason the thermal theory provides no grounds for denying the branched chain character of hydrogen explosion at the upper explosion limit. The complementary chain-carriers theory suggests in explicit terms that self-ignition occurs when the rate of chain-carriers production exceeds the rate of chain-carriers removal. Azatyan [2] showed that when the chain-carries principle is introduced into the classical thermal explosion criterion of Frank-Kamenetskii, the upper explosion limit takes the form of $p^2 = A_u^2 \cdot T_{ig,u}^2 \cdot \exp(\overline{E}_{a,u}/\overline{R}T_{ig,u})$, where P is the pressure, A_u is a fuel dependent constant, $T_{ig,u}$ is the upper ignition temperature, $\overline{R}$ is the universal gas constant, and $\overline{E}_{a,u}$ is the activation energy. Rearrangement results in:

$$\ln(P) = \ln\left(A_u T_{ig,u}\right) + \frac{\overline{E}_{a,u}}{2\overline{R}T_{ig,u}} \tag{1}$$

Liu and Pei [3] considered the intermediate ignition mechanism and showed that it occurs when:

$$2B_1 \exp\left(\frac{-\overline{E}_1}{\overline{R}T_{ig,i}}\right) = \frac{Z_5 P}{\overline{R}T_{ig,i}} B_5 T_{ig,i}^{-0.8} \exp\left(\frac{-\overline{E}_5}{\overline{R}T_{ig,i}}\right) \tag{2}$$

where $T_{ig,i}$ is the intermediate ignition temperature. The constants B and E are respectively the pre-exponential and activation energies of the two elementary competing reactions: 1) $H + O_2 \rightarrow O + OH$ and 5) $H + O_2 + M \rightarrow HO_2 + M$. Here $\overline{E}_1 = 70.30 * 10^3$ kJ/mol, $\overline{E}_5 = 0$, $B_1 = 2*10^{14}$ and $B_5 = 2.3*10^{18}$. Liu and Pei [3] suggested that for hydrogen/oxygen stoichiometric mixture, $Z_5 = 0.8$. We define $A_i = 2\overline{R}B_1/Z_5B_5$ and $\overline{E}_{a,i} = \overline{E}_1$, then:

$$\ln P = \ln\left(A_i T_{ig,i}^{1.8}\right) - \frac{\overline{E}_{a,i}}{\overline{R}T_{ig,i}} \tag{3}$$

Of practical interest is the case when a combustible gas consists of several components. Le-Chatelier suggested that when the combustible properties of each gas are additive, the ignition limit of the mixture correlates well with a simple "parallel-flow" rule [4]. Based on this simple idea, it is thus possible to evaluate the resultant explosion limit temperature, T_{ig}, due to the two ignition mechanisms by knowing the corresponding ignition explosion limit of each. For this task we assume that the rate of chain-carrier production through the upper limit mechanism at a temperature T is proportional to $\sim\exp(-\overline{E}_{a,u}/\overline{R}T)$, and the relative amount of chain-carrier production for $T < T_{ig,u}$ is $\exp(-\overline{E}_{a,u}/\overline{R}T)/\exp(-\overline{E}_{a,u}/\overline{R}T_{ig,u})$. We now approximate the exponential behaviour to an appropriate power, m, and obtain $\exp(-\overline{E}_{a,u}/\overline{R}T)/\exp(-\overline{E}_{a,u}/\overline{R}T_{ig,u}) \approx (T/T_{ig,u})^m$. Similarly we obtain the relative amount of chain-carrier production through the intermediate ignition mechanism $(T/T_{ig,i})^m$.

Following Le-Chatelier rule we obtain the ignition temperature of the mixture, T_{ig}, from $1 = (T_{ig}/T_{ig,u})^m + (T_{ig}/T_{ig,i})^m$, thus:

$$\frac{1}{T_{ig}^m} = \frac{1}{T_{ig,u}^m} + \frac{1}{T_{ig,i}^m} \tag{4}$$

In the present case the two mechanisms through which a chain-carrier is accumulated are not totally independent and thus the parameter, m, is left to be calibrated with experimental observations.

Solution

Eqs. (1) and (3) may be solved to yield explicit terms for $T_{ig,u}$ and $T_{ig,i}$ by using the Lambert function as follows:

$$T_{ig,u} = \frac{-\overline{E}_{a,u}/\overline{R}}{2 \cdot W_{-1}\left[-\frac{\overline{E}_{a,u}}{2\cdot\overline{R}}\frac{A_u}{P}\right]} \tag{5}$$

and,

$$T_{ig,i} = \frac{\overline{E}_{a,i}/\overline{R}}{1.8 \cdot W_0\left[\frac{\overline{E}_{a,i}}{1.8\cdot\overline{R}}\left(\frac{A_i}{P}\right)^{\frac{1}{1.8}}\right]} \tag{6}$$

Here W_j denotes The Lambert function, where the j indices "0" and "−1" indicate the principle and the lower branches, respectively, of the Lambert function that correspond to each of the interval range of the function argument (the term in the brackets).

When we approximate the exponential behaviour of the last term in eq. (1) to an appropriate power m_u we may obtain an approximate expression for $T_{ig,u}: P/A_u T_{ig,u} = \exp(+\overline{E}_{a,u}/2\overline{R}T_{ig,u}) \approx (+\overline{E}_{a,u}/2\overline{R}T_{ig,u})^{m_u}$ and thus $T_{ig,u} = [(+\overline{E}_{a,u}/2\overline{R})^{m_t} A_u/P]^{1/m_u - 1}$, or alternatively,

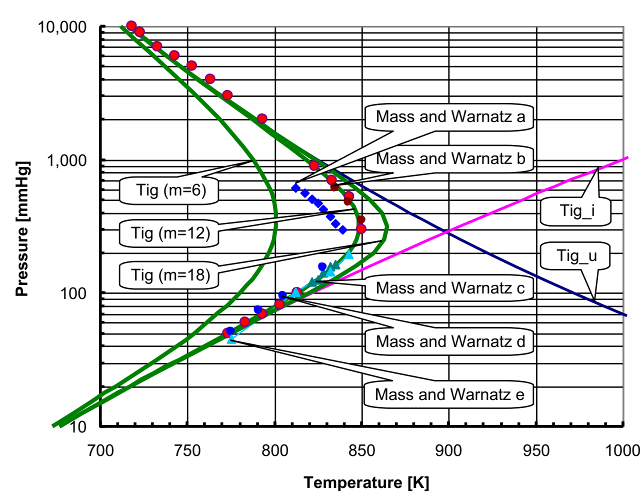

Fig. 1 – Our results vs. various experimental studies. Here T_{ig_u} (eq. (5)), T_{ig_i} (eq. (7)), T_{ig} (eq. (8)). Experimental values: Mass and Warnatz *a*,*b* and *c* for thinly, heavily and KCl coated vessel [7], Mass and Warnatz *d* for KCl coated vessel [8], and Mass and Warnatz *e* for B_2O_3 coated vessel [9].

$$T_{ig,u} = \left(\frac{C_t}{P}\right)^{\frac{1}{m_u-1}} \tag{7}$$

Similarly, eq. (3) becomes $T_{ig,i} = [(+\overline{E}_{a,i}/\overline{R})^{-m_i} A_i/P]^{1/-m_i-1.8}$, or alternatively

$$T_{ig,i} = \left(\frac{C_i}{P}\right)^{\frac{-1}{m_i+1.8}} \tag{8}$$

For best fitting with available experimental data (Lewis and Von-Elbe [5], and Mass and Warnatz [6]), we obtain $m_u = 15$, and $m_i = 8$. We note that the powers of interest are $(m_u - 1)$ and $(m_i + 1.8)$, suggesting a representative typical value of 12 ± 2. We substitute eqs. (7) and (8) in eq. (4), while considering an average value for m:

$$\frac{1}{T_{ig}^m} = \left(\frac{P}{C_u}\right)^{\frac{m}{m_u-1}} + \left(\frac{C_i}{P}\right)^{\frac{m}{m_i+1.8}} \tag{9}$$

Alternatively,

$$\frac{1}{(T_{ig}/T_{ref})^m} = \left(\frac{P}{C_u/T_{ref}^{m_u-1}}\right)^{\frac{m}{m_u-1}} + \left(\frac{C_i/T_{ref}^{m_i+1.8}}{P}\right)^{\frac{m}{m_i+1.8}} \tag{10}$$

For $T_{ref} = 800$ K we obtain $C_u/T_{ref}^{m_u-1} = 1705$ mmHg and $C_i/T_{ref}^{m_i+1.8} = 72.56$ mmHg. Eq. (9) evidently shows that while the upper explosion mechanism dominates the upper pressure region, the intermediate explosion mechanism dominates the lower pressure region. We note that the constants C_u and C_i are based on the physics and chemistry fundamentals of the problem. In practical terms, these constants determine the vertical height of the upper and intermediate explosion limits, respectively. Meanwhile, the powers m_u and m_i determine the slope of these limits and are calibrated here with experimental data. The power m appears to be a valuable parameter to calibrate the peak temperature for which the two mechanisms contribute comparable amounts of chain carriers. Fig. 1 shows our results for $m = 6$, 12, and 18 vs. various experimental results.

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